LECTURE 6

SPATIOTEMPORAL REPRESENTATIONS AND NEURAL FIELDS

Geospatial Representation Learning

PRESS → OR SPACE.

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Learning Outcomes

Lecture

- Explain how temporal dynamics can be represented in geospatial machine learning models.
- Describe neural fields and implicit neural representations for continuous spatial and spatiotemporal data.
- Compare grid-based, sequence-based, graph-based, and coordinate-based representations of Earth system data.
- Assess neural fields for interpolation, compression, forecasting, and multimodal integration.

Lab

- Prepare a small spatiotemporal geospatial dataset for representation learning.
- Implement or use a simple coordinate-based neural representation.
- Evaluate interpolation or reconstruction quality across space and/or time.
- Analyze continuous neural representations compared to discrete raster-based approaches.

PRACTICAL 6

LEARNING A SPATIOTEMPORAL REPRESENTATION

Geospatial Representation Learning

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